



Politechnika Krakowska
im. Tadeusza Kościuszki

Application layer protocols

dr inż. Paweł Król
Katedra Automatyki i Informatyki (E-1)
Politechnika Krakowska

The top layers of the OSI model

	MODEL OSI	PDU	Adres	Urządzenie	Protokoły	MODEL TCP
Implementacja programowa sprzętowa	7 Application				HTTP, DNS, SMTP, POP, SSL, SQL, DHCP	Application
	6 Presentation					
	5 Session					
	4 Transport	Segment TCP Datagram UDP	Port number		TCP, UDP, SCTP	Transport
	3 Network	Packet	IP address	router	IPv4, IPv6, ICMP	Inernet
	2 Data link	Frame	MAC physical address MAC	switch	ARP, Ethernet, PPP, CDP	Network interface
	1 Physical			hub	Bluetooth, DSL, 802.11	

Session layer

Session layer

		MODEL OSI	PDU	Adres	Urządzenie	Protokoły	MODEL TCP
Implementacja	programowa	7 Application				HTTP, DNS, SMTP, POP, SSL, SSL, DHCP	Application
		6 Presentation					
		5 Session					
	sprzętowa	4 Transport	Segment TCP Datagram UDP	Port number		TCP, UDP, SCTP	Transport
		3 Network	Packet	IP address	router	IPv4, IPv6, ICMP	Internet
		2 Data link	Frame	MAC physical address	switch	ARP, Ethernet, PPP, CDP	Network interface
		1 Physical		MAC	hub	Bluetooth, DSL, 802.11	

Session layer

Session Layer Tasks:

- mechanism for opening, closing and managing sessions between end-user application processes
- Cookies
- Example protocol:
 - Remote Procedure Call (RPC)

Presentation layer

Network protocols

		MODEL OSI	PDU	Addres	Device	Protocols	MODEL TCP
Implementation	Software	7 Application				HTTP, DNS	
		6 Presentation				SMTP, POP,	Application
		5 Session				SSL, SQL, DHCP	
	hardware	4 Transport	Segment TCP Datagram UDP	Port Number		TCP, UDP, SCTP	Transport
		3 Network	Packet	IP address	Router (Ruter)	IPv4, IPv6, ICMP	Internet
		2 Link layer	Frame	Physical address MAC	Switch (Przełącznik)	ARP, Ethernet, PPP, CDP	Network Access
		1 Physical			Hub (Koncentrator)	Bluetooth, DSL, 802.11	

Presentation layer

- processing data in a way that is understandable to the application layer
- Data utilizing tasks:
 - formatting
 - compression,
 - encryption

SSL/TLS – basic information

SSL (Secure Sockets Layer):

- Presentation Layer Protocols,
- Secures Application Layer Protocols,
- Symmetric and Asymmetric Keys

TLS (Transport Layer Security):

- Improved version of SSL
- Kryptografia RC4, AES, inne

Version	Publication	Status	Described
SSL 1	1995	Not used	
SSL 2	1995	do 2011	
SSL 3	1996	do 2015	
TLS 1.0	1999	do 2021	RFC 2246
TLS 1.1	2006	do 2021	RFC 4346
TLS 1.2	2008	W użyciu	RFC 5246
TLS 1.3	2018	W użyciu	RFC 8446

Application layer network protocols

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Examples of protocols in application layer

- **DHCP** Dynamic Host Configuration Protocol - UDP port 67
- **DNS** Domain Name Server - TCP / UDP port 53
- **SMTP** Simple Mail Transfer Protocol - TCP port 25
- **POP Post Office Protocol** – TCP port 110
- **IMAP** Internet Message Access Protocol – TCP port 143
- **FTP** File Transfer Protocol – TCP ports 20 i 21
- **SSH** Secure Shell – TCP port 22
- **TELNET** – TCP ports 23
- **HTTP** Hypertext Transfer Protocol – TCP port 80,
- **HTTPS** Hypertext Transfer Protocol Secure – TCP port 443
- and other...

DHCP

Previously discussed...

DNS

DNS (Domain Name Server)

DNS Domain Name Server – TCP/UDP port 53

This address hides IP
149.156.132.5



IP Address	WWW
212.77.98.9	www.wp.pl
149.156.132.5	www.pk.edu.pl
172.217.9.227	www.google.pl
31.13.71.36	www.facebook.com

DNS

INTERNET

What is the address of the website `www.pk.edu.pl`?

The website `www.pk.edu.pl` operates under `149.156.132.5`

DNS server hierarchy

DNS client Local DNS
database Based on
history



Second level domain
servers

www.cisco.com



Root level domain
servers

.pl

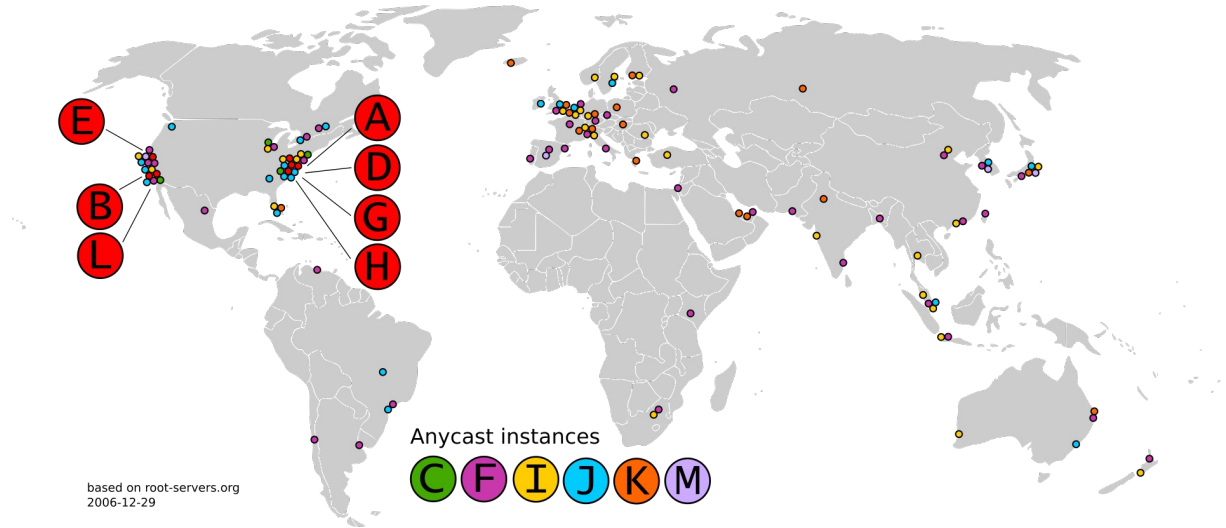
.com



Top-level DNS
servers



ROOT servers



<https://www.iana.org/domains/root/servers>
https://en.wikipedia.org/wiki/Root_name_server

ROOT servers

HOSTNAME	IP ADDRESSES	OPERATOR
a.root-servers.net	198.41.0.4, 2001:503:ba3e::2:30	Verisign, Inc.
b.root-servers.net	199.9.14.201, 2001:500:200::b	University of Southern California, Information Sciences Institute
c.root-servers.net	192.33.4.12, 2001:500:2::c	Cogent Communications
d.root-servers.net	199.7.91.13, 2001:500:2d::d	University of Maryland
e.root-servers.net	192.203.230.10, 2001:500:a8::e	NASA (Ames Research Center)
f.root-servers.net	192.5.5.241, 2001:500:2f::f	Internet Systems Consortium, Inc.
g.root-servers.net	192.112.36.4, 2001:500:12::d0d	US Department of Defense (NIC)
h.root-servers.net	198.97.190.53, 2001:500:1::53	US Army (Research Lab)
i.root-servers.net	192.36.148.17, 2001:7fe::53	Netnod
j.root-servers.net	192.58.128.30, 2001:503:c27::2:30	Verisign, Inc.
k.root-servers.net	193.0.14.129, 2001:7fd::1	RIPE NCC
l.root-servers.net	199.7.83.42, 2001:500:9f::42	ICANN
m.root-servers.net	202.12.27.33, 2001:dc3::35	WIDE Project

<https://www.iana.org/domains/root/servers>

DNS record types

- A
- AAAA
- CNAME
- MX
- NS
- PTR
- SRV
- SOA
- TXT
- CAA
- DS
- DNSKEY

DNS record types

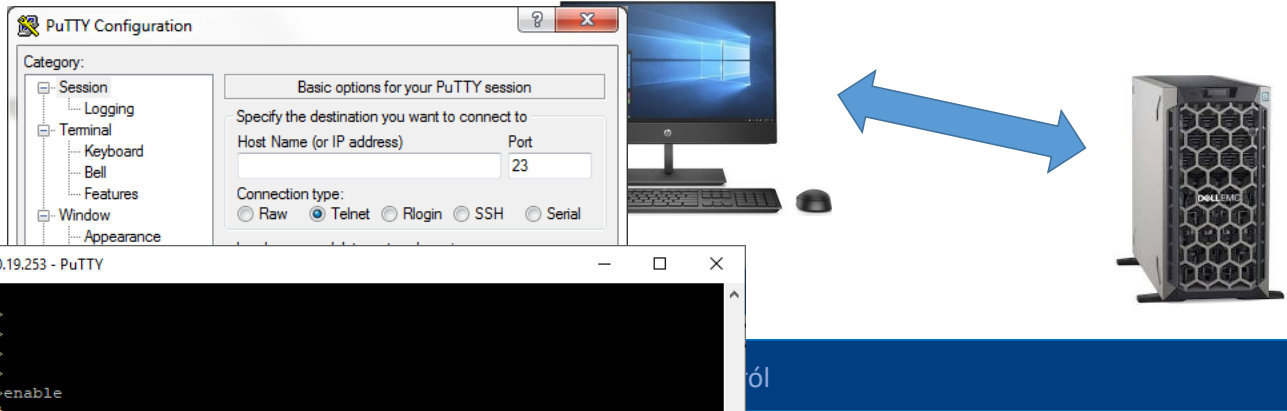
Type	Nr	Description
A	1	Address record – Returns a 32-bit IPv4 address, most commonly used to map hostnames to an IP address of the host, but it is also used for DNSBLs, storing subnet masks in RFC 1101, etc.
AAAA	28	IPv6 address record – returns a 128-bit IPv6 address, most used to map hostnames to IP address of host.
CERT	37	Certificate record – Stores PKIX, SPKI, PGP, etc.
CNAME	5	Canonical name record – Alias of one name to another: the DNS lookup will continue by retrying the lookup with the new name.
MX	15	Mail exchange record – Maps a domain name to a list of message transfer agents for that domain
NS	2	Name server record – Delegates a DNS zone to use the given authoritative name servers
PTR	12	PTR Resource Record – points the IPv4 or IPv6 address to its machine's hostname. It provides a reverse DNS record (rDNS record), by pointing an IP address to the server's hostname.
SOA	6	Start of Authority records – provides essential information about the domain: domain authoritative nameserver, email of the domain administrator, the serial number of DNS zone, etc.
SRV	33	Service record – indicates which specific services the domain operates along with port numbers..
TXT	16	Text record – riginally for arbitrary human-readable text in a DNS record. Allows the website's administrator to insert any arbitrary text in the DNS record.

TELNET/SSH

TELNET

TELNET – TCP port 23

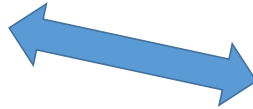
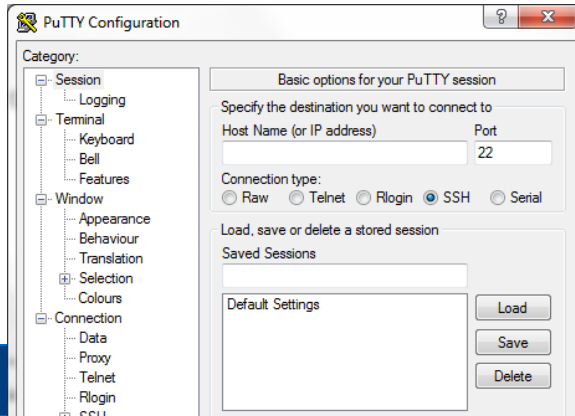
- connection unencrypted
- the ability to eavesdrop on communication



SSH Secure Shell

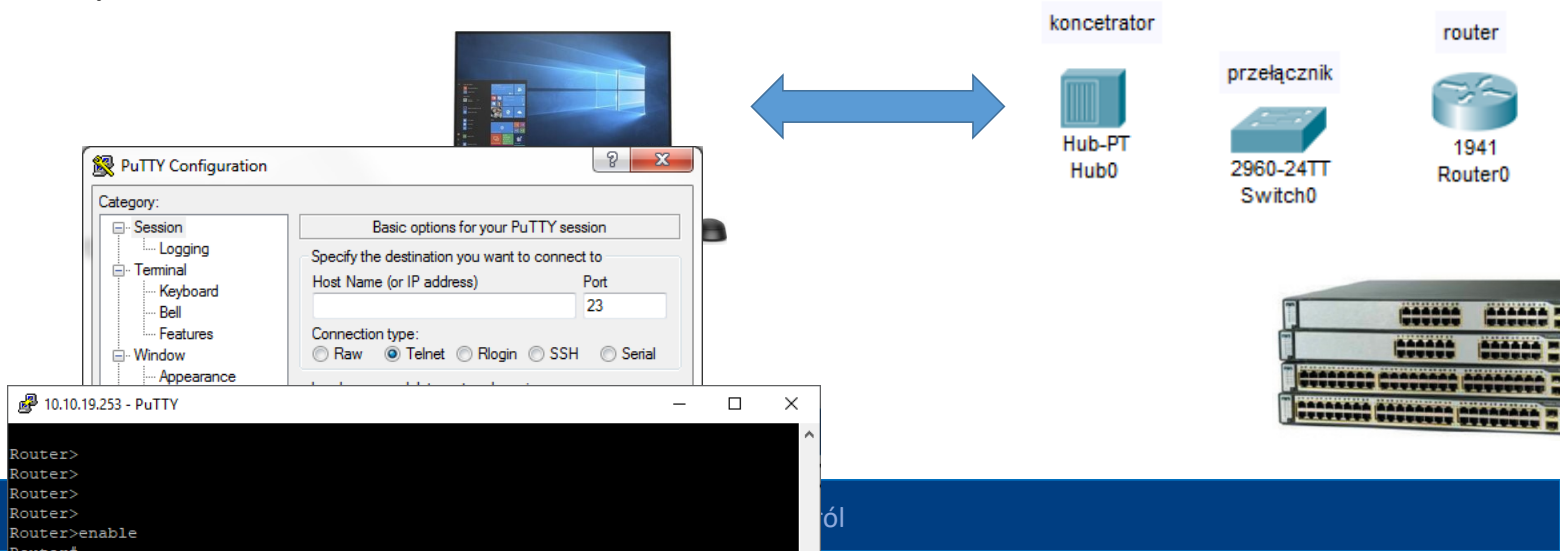
SSH Secure Shell – TCP port 22

- password-encrypted connection (usually AES)
- protection against eavesdropping by third parties



TELNET/SSH and CISCO devices

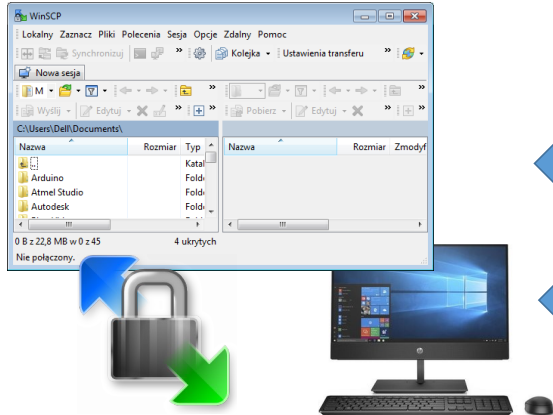
CISCO network devices are programmed remotely via TELNET/SSH protocols



Data Exchange Protocols

FTP (File Transfer Protocol)

FTP CLIENT



FTP SERVER



Control connection

port 21

Uploading files

port 20

FTPs

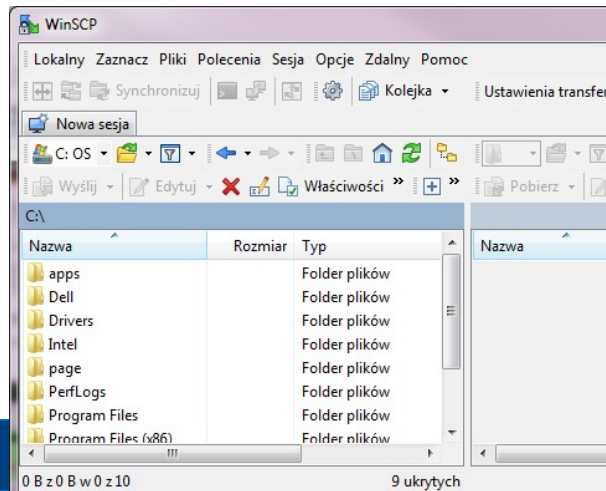
FTP Secure (FTP-SSL)

- encrypted Transport Layer Security (TLS) and Secure Sockets Layer (SSL) protocols,
- uses 990 TCP port for transfer control and 989 TCP for active mode data transfer.

SCP

SCP (Secure copy)

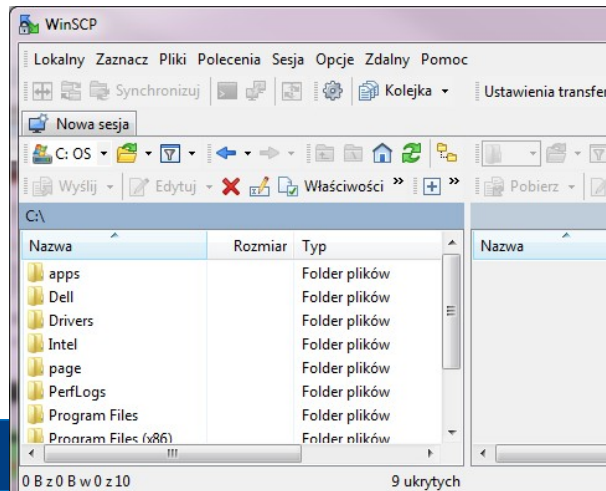
- TCP port 22
- only for file transfer
- it cannot display remote directories or delete files as SFTP does



SFTP

SFTP (ang. SSH File Transfer Protocol) – transferring files using SSH

- connection encrypted as in SSH (Port number 22)
- SFTP is powerful and allows interactive commands to do things like create directories, delete directories, etc.
- sample application: winSCP



E-MAIL

History of e-mail

- 1971 – Ray Tomlinson sent the first email
- 1977 – development of the e-mail specification that is still used today
- 1983 – TCP / IP protocols were used in the new ARPANET network
- 1984 – publication of the DNS specification
- 1979 – Tim Berners Lee at CERN creates the first web project

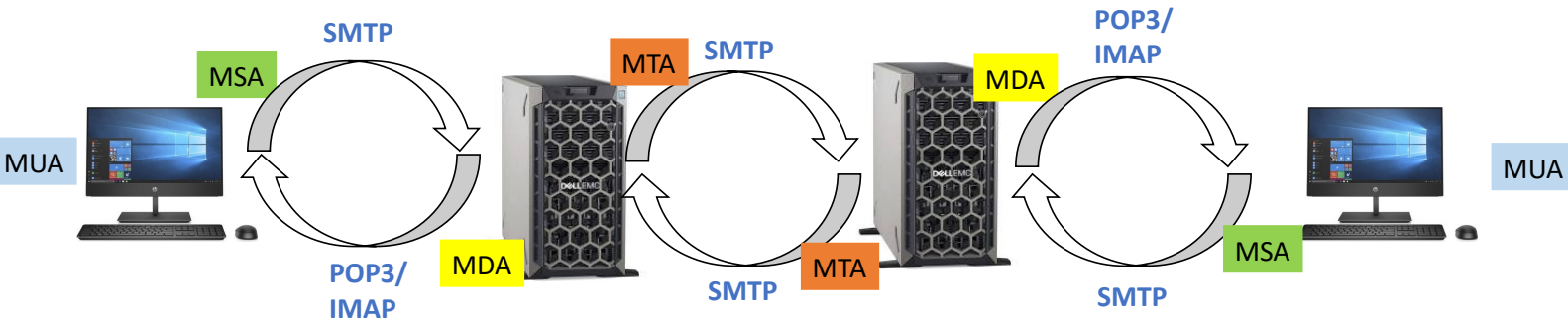
The process of exchanging e-mail

Component of e-mail exchange :

- **MUA Mail User Agent** – an application that receives and sends messages, a program (MS Outlook, Thunderbird) or in a browser,
- **MSA Mail Submission Agent** – a program that handles data from the MUA, validates and submits to the MTA,
- **MTA Mail Transfer Agent** – a server program that receives messages from the MSA or other MTA and then forwards to the MTA or MDA,
- **MDA Mail Delivery Agent** – a server application that collects messages from the MTA and collects them in your mailbox.

The process of exchanging e-mail

- **SMTP** Simple Mail Transfer Protocol – TCP port 25
- **POP** Post Office Protocol – TCP port 110
- **IMAP** Internet Message Access Protocol – TCP port 143



Network protocols used

- **SMTP (ang. Simple Mail Transfer Protocol)** – a communication protocol that describes the method of transmitting electronic mail on the Internet. It transfers data from the computer to the server and between servers.
- **POP3 (ang. Post Office Protocol ver3)** – an internet protocol from the application layer that allows e-mail to be received from a remote server to a local computer via a TCP/IP connection. It only allows you to download and delete mail.
- **IMAP (ang. Internet Message Access Protocol)** – designed as a successor to POP3, it allows you to manage multiple mail folders as well as download and manipulate lists located on a remote server.

SMTP

Ports:

- 25 – not used
- 465
- 587 – default port
- 2525

POP

POP1, POP2

- It is based on the use of SMTP to send messages from the server to the client.

POP3

- Doesn't sync
 - Only retrieves e-mails from the server's inbox.
 - Does not download other folders and their contents (trash, sketches).
- Deletes emails in the sever after download – only one client needed.

Ports: 110 (uses the port), 995

IMAP

IMAP

- It allows you to display e-mails from the server on different devices.
- It synchronizes the entire contents of the mailbox on the server.
- By default, it does not store content on the hard drive.

Ports:

- 143 – unencrypted connections
- 993 – encrypted connections

POP3 vs IMAP

POP3

- + save space for the remote mail-box
- downloading the entire message leaves you vulnerable to viruses

IMAP

- + ability to sync all remote folders across multiple devices

Email content

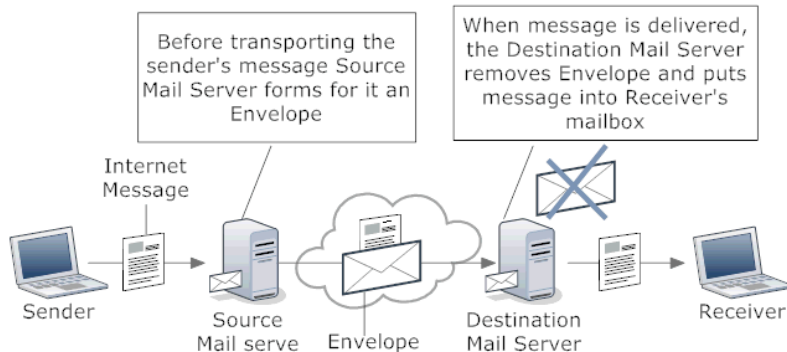
HEADER >>

FROM:
TO:
SUBJECT:

Body >>

Message
Here
has
yyped
contents

The upload process



HTTP

HTTP – URL address

<https://www.pk.edu.pl/index.php?lang=pl>

PROTOCOL

HOST
NAME

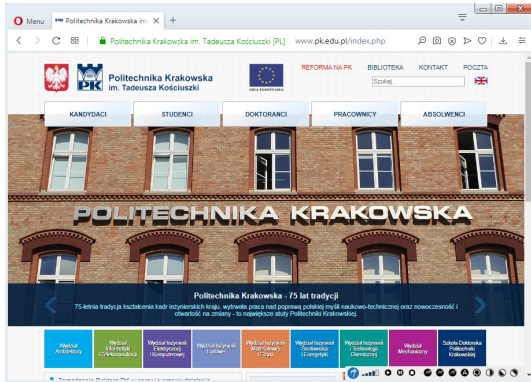
DOMAIN
NAME

PATH

PARAMETERS

HTTP (Hypertext Transfer Protocol)

HTTP – TCP port 80



user

Request
GET

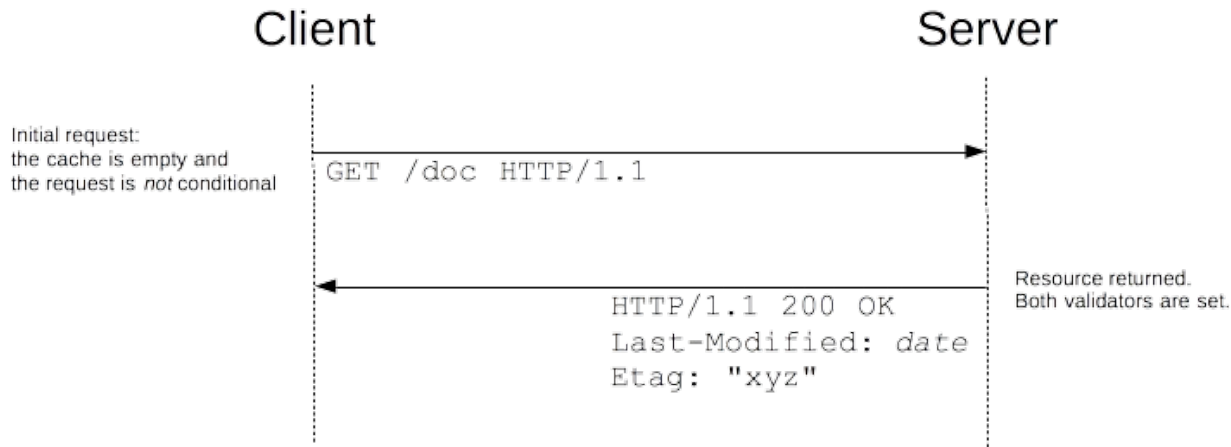


Response

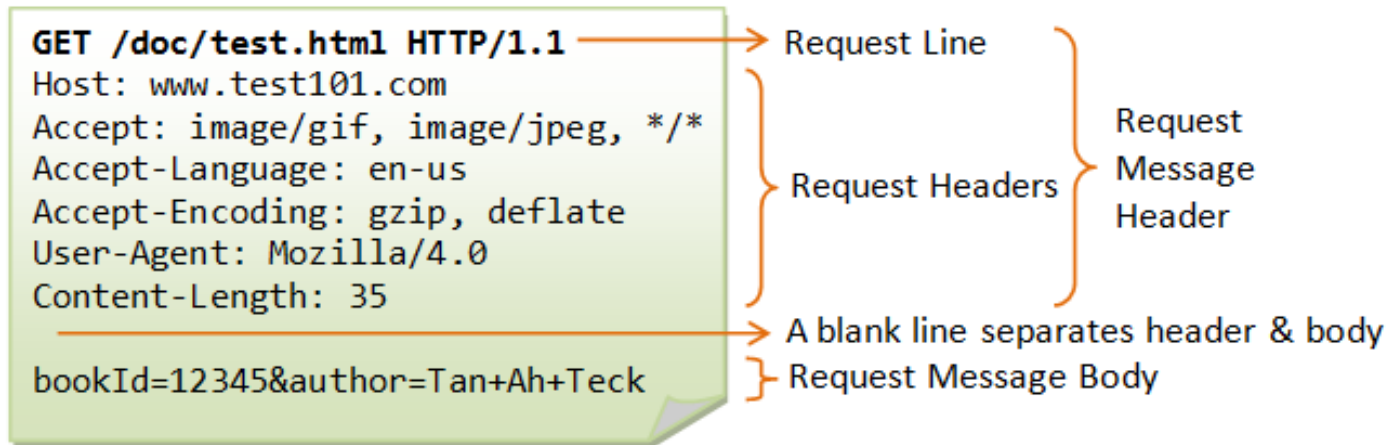


server

HTTP – the format of the question and answer



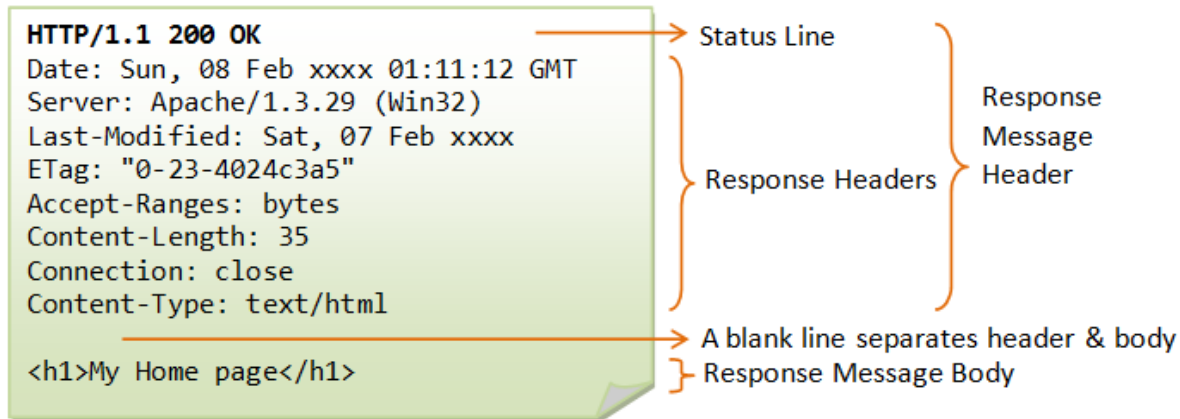
HTTP – query format



HTTP

- GET – requesting data from a specific resource
- POST – sending data to the server to create a resource
- PUT – sending data to the server to update the resource
- DELETE – removal of a specific resource
- PATCH – sending a modification of a resource fragment

HTTP – response format



HTTP – response statuses

The most popular HTTP response codes :

- 200 – OK
- 201 – OK created
- 301 – moved to new url
- 304 – not modified
- 400 – bad request
- 401 – unauthorised
- 404 – not found
- 500 – internal server error

HTTPS

HTTPS Hypertext Transfer Protocol Secure – TCP port 443

The transmitted data is encrypted with protocols from the SSL group
Currently, TLS (SSL 3.1) is used

